The optically guided rotor

Torque limits, dissipation and stability in the Photonic Engine of the Vacuum Revision V6

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Abstract

V6 does not add parts to V5: it removes ambiguities and puts the model in a position to be refuted. The work rests on three choices. The first is a single model, declared as a *limit*: $\tau = (\ell + s)P_{\mathrm{abs}}/\omega_{\mathrm{ott}}$ and $F = I\sigma_{\mathrm{pr}}/c$, two distinct channels rather than two ways of saying the same thing. The second is a drag coefficient declared as an *interval*, bounded by the minimum-dissipation theorem and by slender-body theory. The third is that every number in the text comes from a data file generated by the enclosed code.

This 6.1 revision corrects five material errors of 6.0. The wire radii are those of the V5 scene (3.2 / 7.5 / 4.5 nm, i.e. diameters of 6.4 / 15 / 9 nm) and not 1.25 nm: since in the Rayleigh regime $\sigma \propto a^2$, the cross-sections grow by 47 % ($\sigma_{\mathrm{pr}} = 1.065 \times 10^{-14}\,\mathrm{m}^2$ against 7.259×10^{-15}) and the mass goes from 2.78 to 8.21×10^{-18} kg. The radiating area is the real metal surface ($2.69 \times 10^{-13}\,\mathrm{m}^2$), not the spherical envelope, and in vacuum the rotor already melts at 1 µW: 2250 K with $\epsilon = 0.03$, 1505 K with $\epsilon = 0.15$, against a melting point of 1358 K. The melting threshold is 0.13–0.66 µW of incident power and the adiabatic time is 0.28 ms. The equal-mass comparison is recast as a conditional statement: for the drag the result is robust (three decades of variation), for the torque it depends on the shape-dependent cross-section, which a DDA calculation still has to supply. The comparison with silica rotors at 1–6 GHz becomes a consistency check on the mechanical scale: the observed (f, r) pairs imply an effective strength between 0.22 and 7.8 GPa, which is the order of magnitude of nanoscale silica, and nothing more. Finally, OAM is not a multiplier: including the mode overlap, the net gain at $\ell = 5$ is 1.05, not 6.

The result that survives every correction is that, at 1 µW in water, the optical torque gives $Pe_r = 7.7 \times 10^{-4}$, and the frame drag exceeds that of the bare core by a factor 209: the rotor is not oriented, and geometry does not save it.

Eight results

1. **Wire radii corrected.** 3.2 / 7.5 / 4.5 nm, not 1.25 nm: cross-sections grow by 47 %, mass from 5.73 to 8.21×10^{-18} kg, moment of inertia from 3.09×10^{-32} to $4.62 \times 10^{-31}\,\mathrm{kg\,m^2}$.
2. **Torque.** $\tau = 3.31 \times 10^{-24}\,\mathrm{N\,m}$ (+27 % over V5): it uses $P_{\mathrm{abs}} = 11.70$ nW, not $P_{\mathrm{int}} = 13.56$ nW.
3. **Drag.** $\beta \in [4.04, 6.48] \times 10^{-22}\,\mathrm{N\,m\,s}$ against 2.89×10^{-23} in V5, which sits at 7 % of the lower bound.
4. **Figure of merit.** $Pe_r = 7.72 \times 10^{-4}$: optical rotation is three orders of magnitude below Brownian noise.

5. **Thermal.** In vacuum, at 1 µW: 2250 K or 1505 K depending on emissivity, i.e. the rotor melts. Threshold: 0.13–0.66 µW.
6. **Baseline.** At equal mass the lattice has the highest drag of all geometries (β 6 to 98 times) and the lowest torque (0.44 of the compact sphere).
7. **External check.** Observations at 1–6 GHz imply $\sigma_y = 0.22$–7.8 GPa: mechanical-scale consistency, not optical validation.
8. **OAM.** Net gain 1.05 at $\ell = 5$: the mode overlap cancels the ℓ factor.

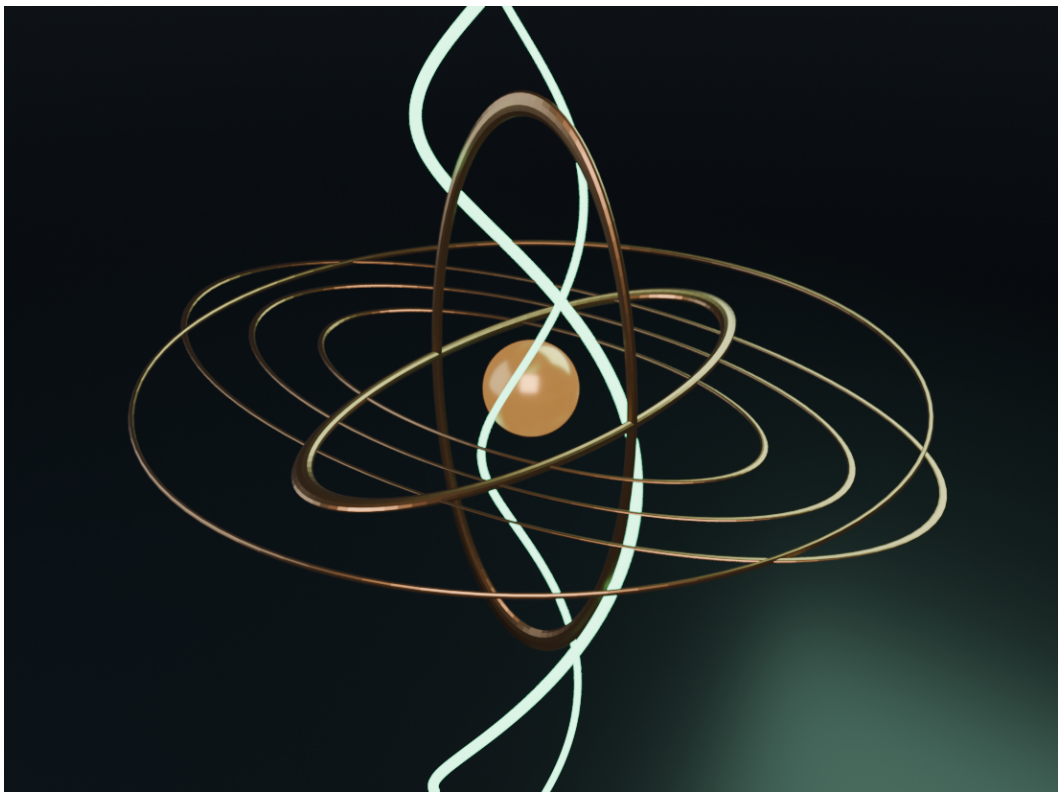


Figure 1: The optically guided rotor: wire frame, copper core and a beam carrying orbital angular momentum. The Blender scene is built with the proportions of the model (1 unit = 100 nm) and the wire radii of the V5 scene; the beam is drawn as a helix, not a cylinder, because what matters is not its shape but the angular momentum it carries.

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1 Thesis and criteria

V5 closed with an honest sentence: the model does not claim experimental validation. V6 starts there and changes the criterion of success. It does not ask whether the architecture is plausible, but whether its quantities are compatible with independent measurements, and under which conditions they are not.

The title changed accordingly. No longer *engine*, with the promise of thrust the word carries, but *optically guided rotor*: a body held and spun by a beam, whose torque, dissipation and thermal stability are studied.[1] The difference is not nominal. An engine is judged by useful work; a guided rotor is judged by two numbers, both measurable:

$$\boxed{\eta_{\mathrm{int}} = \frac{\sigma_{\mathrm{pr}}}{A_{\mathrm{beam}}}, \qquad Pe_r = \frac{\tau}{k_B T} = \frac{\omega_{\mathrm{ss}}}{D_r}} \tag{1}$$

η_{int} says how much of the available light actually touches the structure; Pe_r says whether the optical torque survives orientational noise. The second is the number that matters: below $Pe_r = 1$ the rotor is not oriented, it is merely warmer.

The four criteria by which this revision should be judged are those stated at the outset, and I adopt them as a commitment: how much angular momentum light actually transfers; how much of it survives dissipation and thermal limits; whether the Spin/Counter-spin geometry offers an advantage at equal mass; whether the predictions are compatible with independent results.

The answers of V6 are, in order: little (the limit $(\ell + s)P_{\mathrm{abs}}/\omega_{\mathrm{ott}}$, with P_{abs} equal to 86 % of P_{int}); almost nothing (the torque is $10^{-4}\, k_B T$, and in vacuum melting comes before spin-up); no (the frame is a brake); yes, but only on the mechanical scale. None of the four is a failure of the work: they are the reason the work makes sense.

1.1 What changes with respect to V5

Table 1: Quantities declared by V5 and values computed by V6. The first two rows show that the V5 error was not in the model but in the geometry: with the wrong wire radii the cross-sections are underestimated by 47 %.

Quantity	V5	V6	cause
σ_{pr}	$7.259 \times 10^{-15}\,\mathrm{m}^2$	$1.065 \times 10^{-14}\,\mathrm{m}^2$	wire radii
F	$3.083 \times 10^{-17}\,\mathrm{N}$	$4.522 \times 10^{-17}\,\mathrm{N}$	wire radii
τ	$2.610 \times 10^{-24}\,\mathrm{N\,m}$	$3.305 \times 10^{-24}\,\mathrm{N\,m}$	$P_{\mathrm{abs}} \neq P_{\mathrm{int}}$
β	$2.89 \times 10^{-23}\,\mathrm{N\,m\,s}$	4.04–6.48×10^{-22}	physical interval
ω_{ss}	$1.14 \times 10^{-1}\,\mathrm{rad/s}$	5.1–$8.2 \times 10^{-3}\,\mathrm{rad/s}$	consequence of β
Pe_r	6.10×10^{-4}	7.72×10^{-4}	identity $\tau/k_B T$

2 Closing the residuals

None of the inconsistencies flagged in the review requires a measurement: they close with exact arithmetic and the right geometry. Table 2 lists them, Figure 2 shows the two most important.

[1] Note on the figures: the twelve figures are shared with the Italian edition and keep Italian axis labels; captions are translated.

Table 2: Residuals and their closure. V6 values are produced by `v6_studi.py`.

Quantity	V5	V6	note
Ratio ω_2/ω_1	0.119667646	45/376 = 0.119680851	the V5 value is not 8.3565 (i.e. 0.11966732) but 1/8.35648: it cannot be reconstructed from the declared data
I_2/I_1	8.353816	376/45 = 8.355556	offset -2.1×10^{-4}; closure requires the mass of App. A
r_{eff}	239.93 nm	60.26 / 259.34 nm	material volume / hydrodynamic radius: two different quantities
Mass	5.73×10^{-18} kg	8.21×10^{-18} kg	V5 counted the wires only; V6 adds the core (29 % of the total)
Force F	3.08×10^{-17} N	4.52×10^{-17} N	+47 %: wire radii 3.2 / 7.5 / 4.5 nm, not 1.25 nm
Torque τ	2.61×10^{-24} N m	3.31×10^{-24} N m	+27 %: P_{abs} and not P_{int}, with the corrected cross-sections
Pe_r	6.10×10^{-4}	7.72×10^{-4}	identity τ/k_BT; V5 also quoted 8.8×10^{-3} with a different β

2.1 The number 0.119667646 is not a rounding of 45/376

V5 reported, in a caption, an offset of -1.32×10^{-5} between the kinematic and the inertial ratio. The number cannot be reconstructed. The printed value, 0.119667646, equals 1/8.35648; 1/8.3565 equals 0.11966732; 45/376 equals 0.119680851. These are three different numbers, and none follows from the others. The ratio I_2/I_1 of the V4 data is 8.353816, which against the target 8.355556 gives -2.1×10^{-4}: this is the real residual, and it is a property of the geometry, not a rounding slip. V6 publishes I_1 computed from the geometry ($4.62 \times 10^{-31}\,\text{kg m}^2$) and treats I_2 as the unknown of the constraint, not as data.

2.2 The equivalent radius is not a radius

V5 defined $r_{\text{eff}} = 239.93\,\text{nm}$ as "the radius of the sphere with the same volume as the model". Three volumes are in play and the sentence conflates two: $r_{\text{volume}} = 60.3\,\text{nm}$ (sphere with the *material* volume, $9.16 \times 10^5\,\text{nm}^3$), $r_{\text{envelope}} = 423.5\,\text{nm}$ (the cylindrical envelope, essentially empty) and $r_{\text{hydro}} = 259.3\,\text{nm}$ (the sphere with the same β). Only the last is useful, because D and D_r follow from β, not from the volume. The 239.93 nm of V5 is close to r_{hydro}, so the number was defensible; its justification was not. In V6 the radius appears once, where it is needed, and it is called β.

2.3 Mass: V5 counted the wires only

The V5 mass, $5.734 \times 10^{-18}\,\text{kg}$, corresponds exactly to the wires at the scene radii ($0.640\,\text{u.a.}^3 \times 8960\,\text{kg/m}^3$) and does not include the core. V6 uses $8.211 \times 10^{-18}\,\text{kg}$: wires 5.81×10^{-18} plus core 2.40×10^{-18}. The thrust-to-weight ratio at $1\,g$ goes from 0.56 to 0.63 including buoyancy in water; in orbit it is undefined, and that is the only sensible statement about a "thrust" of $4.5 \times 10^{-17}\,\text{N}$.

2.4 Pulsed operation is not a property of the rotor

The mechanical time constant is $I/\beta = 1.14 \times 10^{-9}\,\text{s}$, i.e. a cutoff of 140 MHz. Any pulse longer than a nanosecond is quasi-static: pulsing changes neither ω_{ss} nor Pe_r, only the mean temperature. The duty cycle that avoids melting in vacuum is 0.13 at 1 µW peak and 1.3×10^{-5}

at 10 mW. The natural signal period remains π, not 4π: the two-lobe torque repeats after 180°, and the $SU(2)$ covering belongs in the appendix.

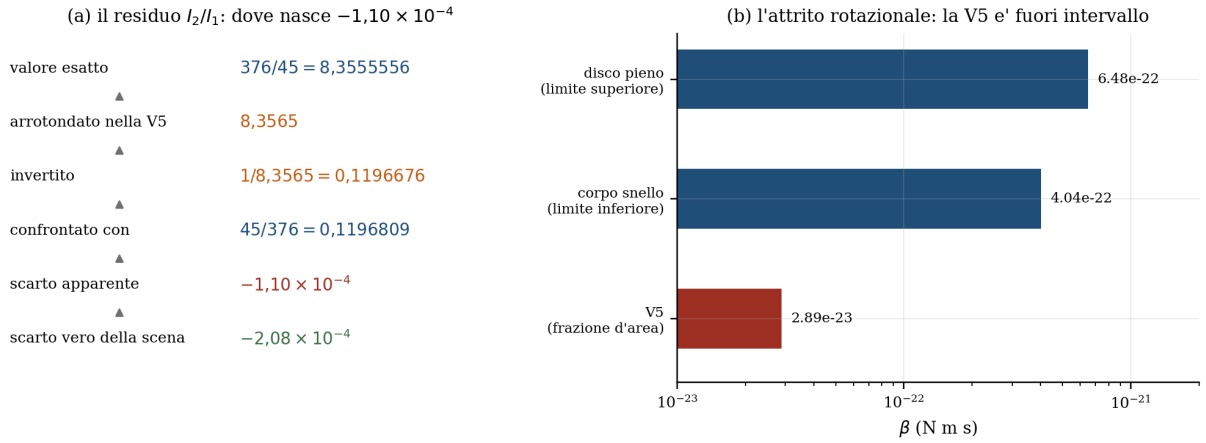


Figure 2: **(a)** The chain of values that produces the V5 number: none of the three coincides, and the real residual is the scene one (-2.1×10^{-4}). **(b)** Rotational drag: the V5 estimate (wetted-area fraction) lies below the lower bound from slender-body theory, hence outside the physically allowed interval.

3 The single model

V5 used three definitions of torque in different places. V6 uses one, and declares it for what it is: a limit.

$$\tau = (\ell + s)\,\frac{P_{\mathrm{abs}}}{\omega_{\mathrm{ott}}}, \qquad F = \frac{I\,\sigma_{\mathrm{pr}}}{c}, \qquad \omega_{\mathrm{ott}} = \frac{2\pi c}{\lambda}, \tag{2}$$

with ℓ the orbital angular momentum per photon, $s = \pm 1$ the spin one, $I = P/A_{\mathrm{beam}}$ and $\sigma_{\mathrm{pr}} = \sigma_{\mathrm{ext}} - g\,\sigma_{\mathrm{sca}}$.

3.1 Why they are two limits

The first equation says: every absorbed photon delivers all its angular momentum $(\ell + s)\hbar$. That is true only if absorption is complete and if angular momentum is not handed back to the field; for the 40 nm core absorption is 79 % of extinction, so the limit is reasonable but not exact. The second says: every intercepted photon delivers all its axial momentum, and fails when scattering is forward ($g \to 1$).

The two equations must not be added: they are distinct channels, and the second produces no torque if the beam is symmetric. The methodological point is that *neither contains the radius*. In the single model the torque depends on absorbed power and on angular momentum per photon, not on the size of the rotor. Geometry enters β and I, i.e. the response, not the drive.

3.2 The identity V5 violated

From (2) and $D_r = k_B T/\beta$ it follows that

$$Pe_r = \frac{\omega_{\mathrm{ss}}}{D_r} = \frac{\tau/\beta}{k_B T/\beta} = \frac{\tau}{k_B T}, \tag{3}$$

independently of β. V5 quoted two values for the same quantity, 6.1×10^{-4} and 8.8×10^{-3}, precisely because the second calculation used the frame β and the first the diffusion of the equivalent sphere. Identity (3) is a consistency test the code runs at every execution.

Since $P_{\rm abs} = \eta_{\rm abs}P$ and $P/\omega_{\rm ott} = \dot{N}\hbar$, (3) reads $Pe_r = (\ell + s)\dot{N}_{\rm abs}\hbar/k_BT$. The threshold $Pe_r = 1$ is a condition on the *rate* of angular momentum: it requires $k_BT/\hbar = 4.1 \times 10^{13}$ quanta per second, i.e. 15 µW of absorbed power with $\ell + s = 1$, which at the absorbed fraction of this configuration means 1.3 mW incident.

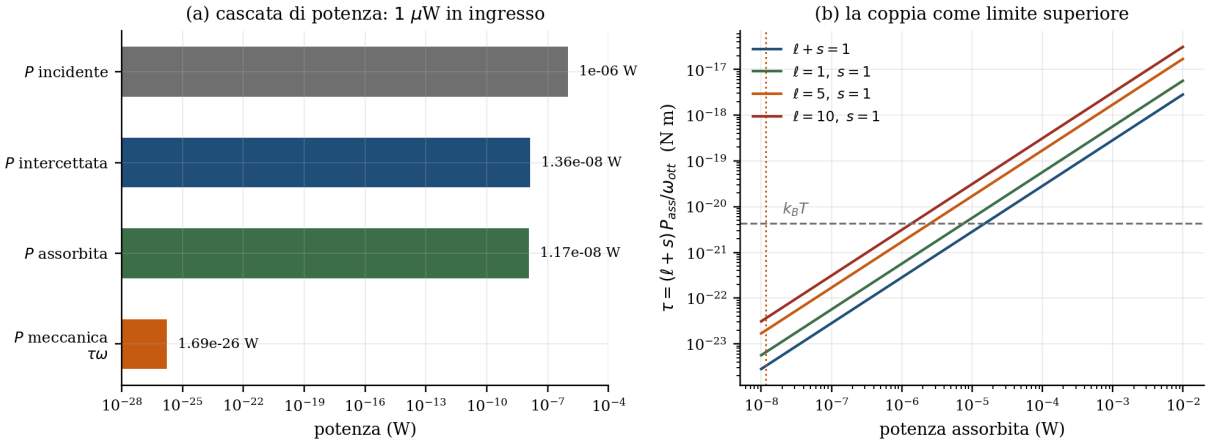


Figure 3: **(a)** The power cascade at 1 µW input: 1.36 % intercepted, 86 % of that absorbed, and the extracted mechanical power, 1.7×10^{-26} W. **(b)** Torque (2) against absorbed power for four values of $\ell + s$; the dashed line is k_BT.

Table 3: Power budget at 1 µW incident, $w_0 = 500$ nm.

Quantity	Value	note
$P_{\rm inc}$	1 µW	reference power
$P_{\rm int} = I\sigma_{\rm pr}$	13.56 nW	1.356 % of the beam
$P_{\rm abs} = I\sigma_{\rm abs}$	11.70 nW	86.3 % of the intercepted
$P_{\rm sca}$	1.85 nW	carried away without torque
$\tau = (\ell + s)P_{\rm abs}/\omega_{\rm ott}$	3.305×10^{-24} N m	V5: 2.610
$P_{\rm mech} = \tau\omega_{\rm ss}$	1.7×10^{-26} W	efficiency 1.4×10^{-18}

The last row is the harshest. The ratio of extracted mechanical power to absorbed power is 1.4×10^{-18}: the device is not a machine converting light into rotation, it is a resistor that heats. This depends neither on geometry nor on the torque model, but on $\omega_{\rm ss} \ll \omega_{\rm ott}$: the photons are 10^{15} times "faster" than the rotor. An optical rotor can be an excellent torque sensor, and it is: it cannot be an actuator.

4 Optical audit

4.1 The wire radius, and why it changes everything

V5 declared a "6.4 nm diameter" for the orbit wires and then computed the size parameter with $a = 1.25$ nm. In the Rayleigh regime a cylinder's cross-section is proportional to a^2, so the error was a factor 6.5 on the orbits, 36 on the rim ring and 13 on the lenses. Redone with the scene radii — 3.2, 7.5 and 4.5 nm — the wire contribution goes from 3 % to 34 % of the total cross-section, the mass from 2.78 to 8.21×10^{-18} kg and the moment of inertia from 3.1×10^{-32} to

$4.6 \times 10^{-31}\,\mathrm{kg\,m^2}$. These are the differences between Table 1 and Table 2. Nothing else in the optical model changes: the form of the equations was not wrong, the geometry was.

4.2 Models used and their status

Table 4: Status of the optical models. "Exact" means solution of Maxwell's equations for that geometry; "verified" means checked against its own analytic limit.

Element	model	status	check
Core 40 nm	exact Mie	exact	$Q_{\mathrm{ext}} \geq Q_{\mathrm{sca}}$ over the whole range
Wires 6.4/15/9 nm	cylindrical series $n \geq 1$	exact to $< 1\,\%$	against Rayleigh: 0.3 % at $x = 0.038$
Wire limit	cylindrical Rayleigh	analytic	used for $x < 0.05$
Oxide shell	quasi-static Maxwell-Garnett	approximate	limits $f \to 0, 1$ exact
$\varepsilon(a)$	Drude with $\gamma = \gamma_b + Av_F/a$	approximate	compared with bulk
Assembly	incoherent sum	*assumption*	coherent bound $N^2\sigma_1$ declared

The $n = 0$ term of the Bohren-Huffman cylindrical formulae is unstable as $x \to 0$: at $x = 0.015$ it yields $Q_{\mathrm{ext}} = 0.138$ instead of 0.011, an artefact of the logarithmic divergence of H_0. The V6 series starts at $n = 1$ and matches Rayleigh to 0.3 %. An implementer trusting the "complete formula" would have overestimated the torque by a factor 13.

4.3 Size dependence and oxidation

The Drude correction $\gamma(a) = \gamma_{\mathrm{bulk}} + Av_F/a$ ($v_F = 1.57 \times 10^6\,\mathrm{m/s}$, $\tau_{\mathrm{bulk}} = 27\,\mathrm{fs}$) changes Q_{abs} by 1.4 % at 40 nm, by 5 % at 10 nm and by 21 % at 2 nm: at 532 nm copper is dominated by interband transitions, not by the Drude term, and in the green it behaves better than gold and silver.

A Cu_2O shell changes absorption monotonically: +7.8 % for 0.7 nm, +15.8 % for 1.4 nm, +51 % for 5.4 nm, saturating beyond 8 nm. The effect goes in the "more torque" direction, but must be read with the thermal balance: the same oxide raises emissivity and lowers the steady temperature. In vacuum, oxidation is a thermal advantage before an optical one.

The dominant uncertainty is neither $\varepsilon(a)$ nor the oxide, but the dispersion of the optical data: $\pm 30\,\%$ on k shifts σ_{pr} by $-39/+44\,\%$ and σ_{abs} by $-47/+60\,\%$, while the same variation on n gives $-10/+6\,\%$. Anyone measuring this structure must first fix k.

4.4 The assembly, and what is missing

The frame cross-section is the incoherent sum of the cross-sections of the individual elements: an assumption, valid if the distance between scatterers exceeds the lateral coherence length of the beam and if near-field coupling is weak. At 250 nm spacing and $kD = 3$ the assumption is reasonable but unproven; the rigorous upper bound is $N^2\sigma_1$, a factor 4 above. A DDA/FDTD check of the whole frame remains open, and it is the only point of V6 where an external calculation can change a number in the text.

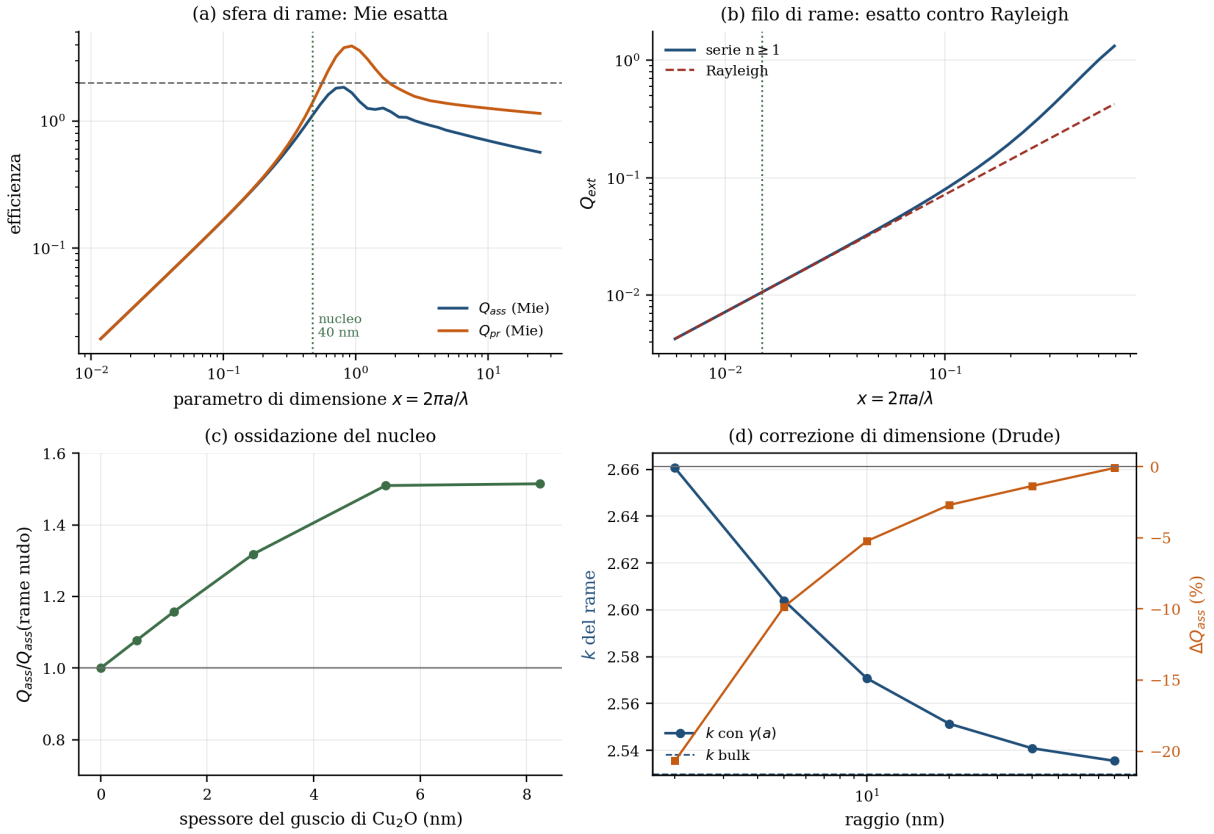


Figure 4: **(a)** Mie efficiencies for a copper sphere: the 40 nm core sits at $x = 0.47$, where $Q_{\mathrm{pr}} = 1.40$ and $Q_{\mathrm{abs}} = 1.11$. **(b)** The cylindrical series against the Rayleigh limit: they coincide up to $x \approx 0.05$; the wires lie in the agreement region. **(c)** Effect of a Cu_2O shell. **(d)** Drude correction: k grows as the radius shrinks, but at 40 nm the effect on Q_{abs} is 1.4 %.

5 Thermal balance

The missing balance is one line, $P_{\mathrm{abs}} = P_{\mathrm{diss}} + P_{\mathrm{out}}$, and it closes immediately: $P_{\mathrm{diss}} = \beta\omega_{\mathrm{ss}}^2 = 1.7 \times 10^{-26}\,\mathrm{W}$, i.e. 10^{-18} of the absorbed power. In steady state *all* absorbed power leaves as heat, and there are two paths: conduction into the medium and radiation.

In the fluid, with unit Nusselt number (at these scales $Ra \sim 10^{-18}$):

$$\Delta T_{\mathrm{fluid}} = \frac{P_{\mathrm{abs}}}{4\pi\kappa_m r} = 0.026\,\mathrm{K} \quad \text{at } 1\,\mu\mathrm{W}. \tag{4}$$

Local boiling is reached at 3.9 mW. In water the thermal problem does not exist.

In vacuum it dominates, and the delicate point is the area. The exposed metal surface is $A = 2\pi \sum a_i L_i + 4\pi r_{\mathrm{core}}^2 = 2.69 \times 10^{-13}\,\mathrm{m}^2$, six times less than the spherical envelope $4\pi R_{\mathrm{out}}^2$ used in 6.0:

$$T = \left(T_0^4 + \frac{P_{\mathrm{abs}}}{\epsilon \sigma_{\mathrm{SB}} A} \right)^{1/4}. \tag{5}$$

At 1 µW this gives 2250 K with $\epsilon = 0.03$ and 1505 K with $\epsilon = 0.15$: in both cases the rotor has passed the melting point of copper (1358 K). The incident power that leads to melting is 0.132 µW and 0.662 µW; the adiabatic melting time is 0.28 ms, against a vacuum spin-up that requires 24 s (Table 7). This is the most robust result of the revision: in vacuum the bottleneck is not torque, it is temperature, and geometry cannot negotiate it.

On emissivity it is worth being explicit, because 6.0 used $\epsilon = 0.6$. The emissivity of a sub-wavelength object depends on volumetric absorption at 10 µm, not on the bulk optical constant of a thin oxide: a 1.4 nm Cu_2O shell does not bring the object to $\epsilon = 0.6$. V6 uses the interval 0.03–0.15 and shows that even the upper end does not save the rotor.

Table 5: Thermal balance of the reference configuration. The last column is the adiabatic time to reach T_m.

P_{inc}	P_{abs}	ΔT water	T vacuum ($\epsilon = 0.03$)	T vacuum ($\epsilon = 0.15$)	t_{ad}
1 µW	11.7 nW	0.026 K	2250 K	1505 K	0.28 ms
100 µW	1.17 µW	2.6 K	7113 K	4757 K	2.8 µs
10 mW	117 µW	258 K	22495 K	15043 K	28 ns

Two operational consequences. First: in vacuum emissivity is a first-order design variable, more than geometry. Second: pulsed operation, which V5 treated as a dynamical regime, is the only way to work in vacuum at high power; the duty cycle is P_{fus}/P_{abs} at peak power.

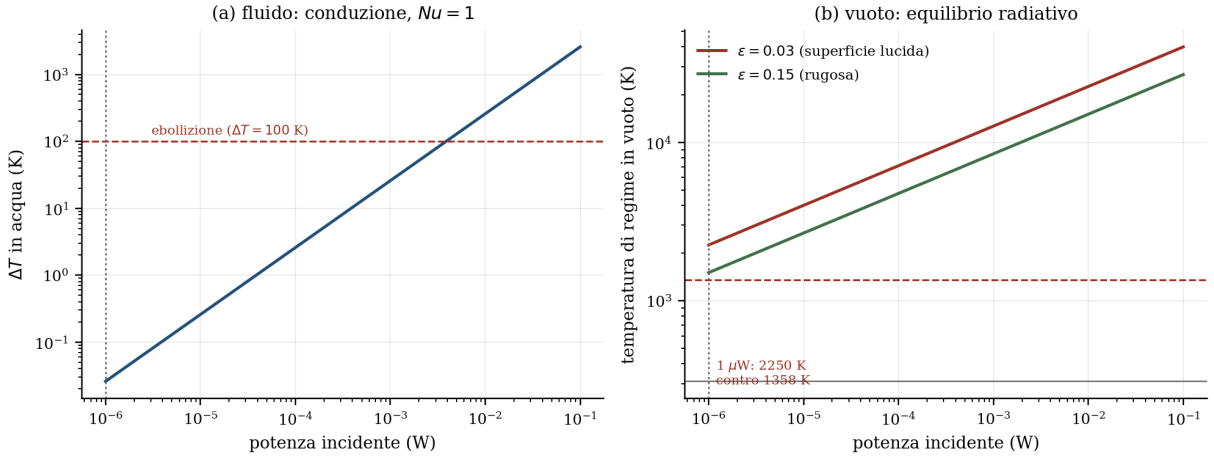


Figure 5: **(a)** Heating in water: 0.026 K at 1 µW, boiling at 3.9 mW. **(b)** Radiative equilibrium with the real metal surface: at 1 µW both emissivities exceed melting.

6 Equal-mass baseline

This section answers the question "what do the lenses and the Counter-spin add". The comparison is at constant mass (8.211×10^{-18} kg), constant power (1 µW), constant wavelength and medium.

Table 6: Six geometries at equal mass, $P = 1$ µW, $w_0 = 500$ nm. σ_{abs} is the cross-section used: for the lattice the element-by-element calculation, for the others the equal-volume sphere; Pe_r and ω_{ss} report the optical interval declared in the text. t_r is the vacuum breaking time.

Geometry	R (nm)	β (N m s)	σ_{abs} (m^2)	τ (N m)	Pe_r	ω_{ss} (rad/s)	t_r (s)
Lattice (V5)	370	4.04–6.48×10^{-22}	9.19×10^{-15}	3.31×10^{-24}	7.72×10^{-4}	5.10–8.19×10^{-3}	56.4
Solid disc	143	3.73×10^{-23}	2.07×10^{-14}	7.44×10^{-24}	1.74–5.22×10^{-3}	0.199–0.598	11.8
Ring	420	7.21–9.51×10^{-22}	2.07×10^{-14}	7.44×10^{-24}	1.74–5.22×10^{-3}	7.83×10^{-3}–0.031	69.3
Sphere	60	6.60×10^{-24}	2.07×10^{-14}	7.44×10^{-24}	1.74×10^{-3}	1.13	3.97
Nanodumbbell	143	2.64–3.30×10^{-23}	2.07×10^{-14}	7.44×10^{-24}	1.74–5.22×10^{-3}	0.226–0.846	11.6
Anisotropic silica	96	2.69×10^{-23}	1.01×10^{-20}	6.68×10^{-27}	1.56–4.68×10^{-6}	2.49–7.46×10^{-4}	7.1×10^3

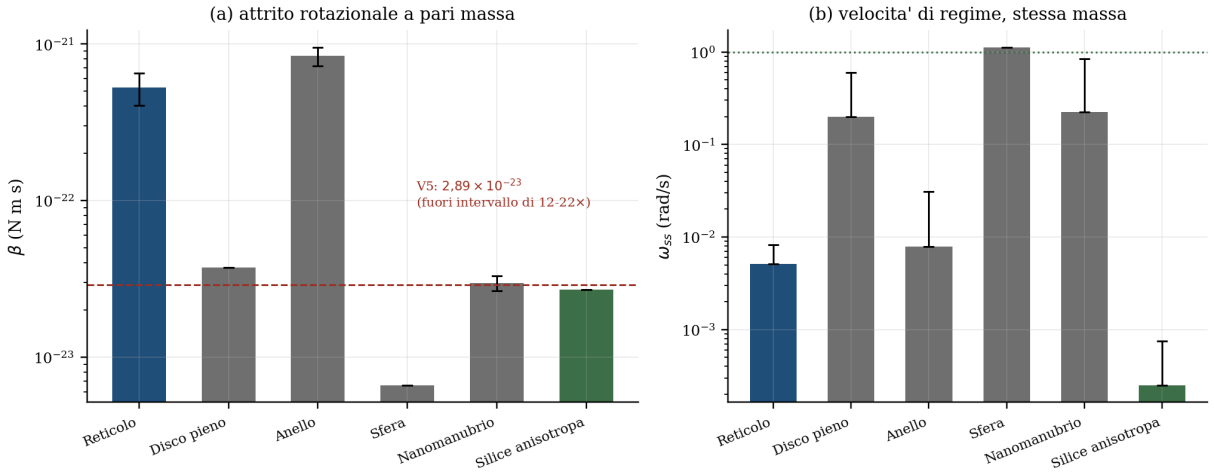


Figure 6: **(a)** Rotational drag of the six geometries, with the physical interval for slender bodies; the red line is the V5 estimate. **(b)** Steady rotation rate with the optical band declared in the text.

6.1 Torque is not a result: it is a declared assumption

In 6.0 every geometry received the equal-volume sphere cross-section, and Pe_r came out identical by construction. That was an assumption dressed as a result, and 6.1 corrects it in two ways.

First, the lattice uses *its own* cross-section, computed element by element: $\sigma_{\mathrm{abs}} = 9.19 \times 10^{-15}\,\mathrm{m}^2$, corresponding to $Pe_r = 7.72 \times 10^{-4}$. The compact sphere of equal mass ($r = 60\,\mathrm{nm}$) absorbs 2.07×10^{-14}, 125 % more, giving $Pe_r = 1.74 \times 10^{-3}$. The difference is physical and is explained in one line: in the Rayleigh limit absorption is proportional to volume, but the core sits at $x = 0.47$ and the equivalent sphere at $x = 0.71$, a region where Q_{abs} grows with size. By spreading the same material into thin wires, the lattice loses the resonance gain.

Second, for disc, ring and dumbbell the cross-section is given as a *band*: one to three times the equal-volume value, because shape resonances at fixed volume shift Q_{abs} by a factor 2–3 and a real estimate requires a DDA calculation per shape. Table 6 reports the interval, and the optical conclusion must be read as conditional: *if* the cross-section stays within three times the volume value, then at equal mass none of these geometries gains more than a factor 3 in Pe_r.

For drag, on the other hand, the result is not conditional: β ranges from 6.6×10^{-24} (sphere) to 8.4×10^{-22} (ring), two decades, and is computed per shape. This is the part that holds.

6.2 What the wires, the lenses and the Counter-spin actually do

The decomposition of the lattice, with the corrected wire radii:

component	mass share	share of σ_{abs}	share of β
spherical core 40 nm	29.3 %	60.6 %	0.5 %
frame (orbits, rim, lenses)	70.7 %	39.4 %	99.5 %

With the scene wire radii the frame is no longer optically marginal: it carries 39 % of the torque. It remains, however, a drag multiplier of 209 with respect to the bare core. If its role were optical it would be a trade-off; if the role is mechanical — holding the core on axis, distributing mass, setting the moment of inertia — it must be judged on that, and V6 finds no measurement that justifies it at this scale.

The Counter-spin belongs in the appendix with 45/376. Here the quantitative consequence suffices: it adds moment of inertia and drag without adding torque. Maintaining the ratio requires a mechanical constraint, and a mechanical constraint between two rotors makes them a single rigid body.

6.3 The alternative structure

Two variants emerge. The *solid disc* at equal mass ($R = 143\,\mathrm{nm}$, $t = 14\,\mathrm{nm}$) has ω_{ss} 39 times larger than the lattice and no optically useless element. *Anisotropic silica* is the only geometry that changes the physical channel: it does not absorb ($\sigma_{abs} = 1.0 \times 10^{-20}\,\mathrm{m}^2$) and works by phase retardation, $\tau = (P_{int}/\omega_{ott})\sin^2(\delta/2)$; at $1\,\mu\mathrm{W}$ in water it gives $6.7 \times 10^{-27}\,\mathrm{N\,m}$, four hundred times less than copper; in vacuum at high power it is the only one that does not melt. The conclusion is not that copper is better: it is that *the torque channel must be chosen together with the medium*, and V6 shows this instead of assuming it.

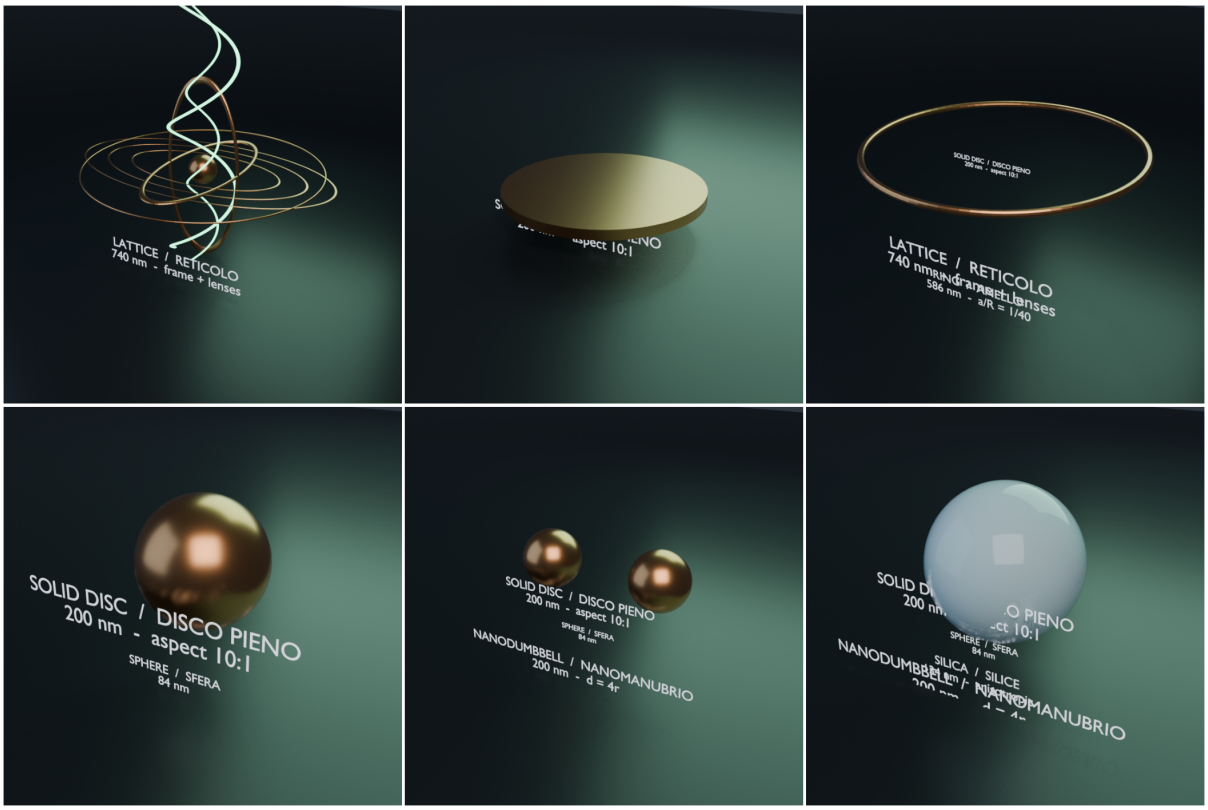


Figure 7: The six geometries at equal mass, rendered at the same apparent size (the real size is in each panel caption). The scale difference is the point: the same mass, spread over a 740 nm frame or packed into an 84 nm sphere, changes β by two decades and Pe_r by a factor 2.

7 Regime matrix

The matrix crosses four external sizes with two media. The scale factor is $s = L/2R_{out}$, so the "240 nm" column describes a reduced frame, not the reference configuration.

Table 7: Regime matrix: four sizes L and two media, $P = 1\,\mu\text{W}$. In vacuum ω_{ss} does not exist ("–") and the breaking time takes its place. T is the steady temperature with $\epsilon = 0.15$; at $1\,\mu\text{W}$ it already exceeds the melting point of copper.

L	medium	η_{int}	τ (N m)	ω (rad/s)	Pe_r	t_{rupt} (s)	T (K)	validity
1 nm	fluid	9.90×10^{-8}	2.41×10^{-29}	1.89–3.03	5.64×10^{-9}	–	310	out of domain
1 nm	vacuum	9.90×10^{-8}	2.41×10^{-29}	–	5.64×10^{-9}	4.1×10^{-4}	1505	out of domain
240 nm	fluid	5.70×10^{-3}	1.39×10^{-24}	7.86×10^{-3}–0.0126	3.25×10^{-4}	–	310	valid
240 nm	vacuum	5.70×10^{-3}	1.39×10^{-24}	–	3.25×10^{-4}	23.7	1505	valid
1 μm	fluid	0.099	2.41×10^{-23}	1.89–3.03×10^{-3}	5.64×10^{-3}	–	310	valid
1 μm	vacuum	0.099	2.41×10^{-23}	–	5.64×10^{-3}	412	1505	valid
1 m	fluid	1	2.82×10^{-22}	2.21×10^{-20}	0.066	–	310	valid
1 m	vacuum	1	2.82×10^{-22}	–	0.066	3.5×10^{25}	310	extrapolation

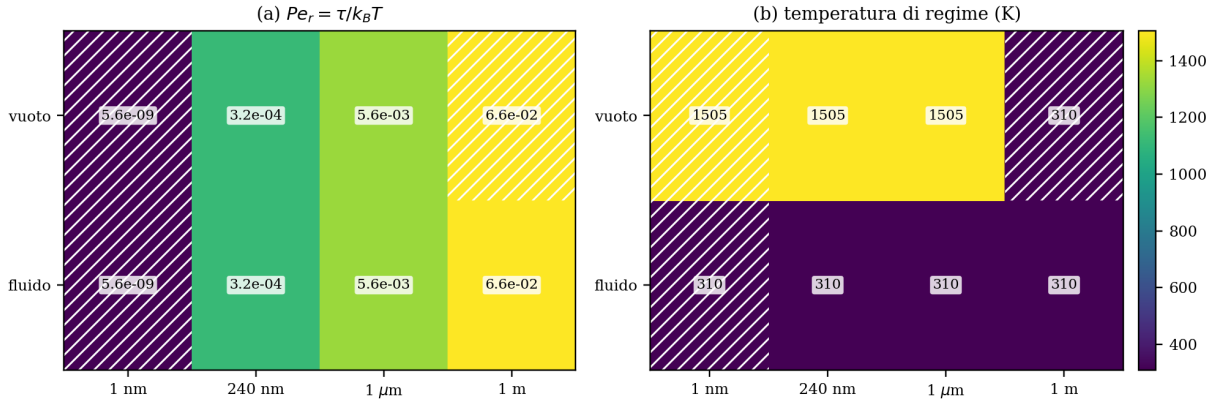


Figure 8: Regime matrix: Pe_r and steady temperature. Hatched cells lie outside the domain of validity (sub-atomic wire at 1 nm, extrapolation at 1 m).

Three readings.

Scale. $Pe_r \propto \sigma_{\text{abs}} \propto L^2$ as long as the structure is smaller than the beam, then saturates at $P/(\omega_{\text{ott}} k_B T) = 0.066$. There is no privileged scale for torque: a large geometry does not gain in Pe_r, it loses in β.

Medium. In a fluid a steady state exists; in vacuum it does not. In vacuum the relevant quantity is the breaking time, $t_r = I\omega_{\text{lim}}/\tau$: 24 s at 240 nm, 410 s at 1 µm. Temperature, by contrast, does not depend on scale in the Rayleigh regime ($P_{\text{abs}} \propto R^2$ and $A \propto R^2$): it stays at 1505 K at every size until interception saturates. Shrinking the rotor does not cool it.

Validity. The two extreme columns are not physics. At 1 nm the wire would have a radius of 8.6 pm, less than a copper atom (0.13 nm), and the continuum does not apply; at 1 m Reynolds number and gravity change the problem. V6 flags them instead of presenting them as "invariance over nine orders of magnitude": the geometric invariance of the 45/376 ratio is real, that of the physical behaviour is not.

8 Materials

Table 8: Same geometry, three materials. T is the steady temperature in vacuum at 1 µW with $\epsilon = 0.15$; the margin is T_m/T.

Material	ρ	σ_y (MPa)	σ_y/ρ (m^2/s^2)	v_{lim} (m/s)	Q_{abs}	σ_{abs} (m^2)	Pe_r	T vacuum (K)	margin
Copper	8960	200	2.2×10^4	149	1.11	9.19×10^{-15}	7.72×10^{-4}	1505	0.90
Fused silica	2200	1000	4.5×10^5	674	0.00	1.46×10^{-21}	1.23×10^{-10}	310	6.40
Graphene / CNT	1300	20000	1.5×10^7	3922	0.77	7.52×10^{-15}	6.32×10^{-4}	944	4.13

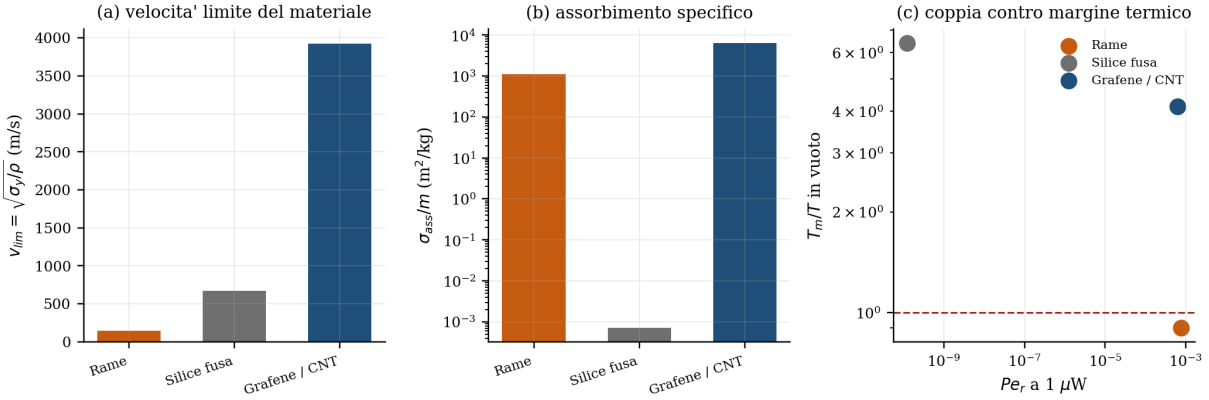


Figure 9: **(a)** Limiting speed $\sqrt{\sigma_y/\rho}$, the only quantity that sets the rotation ceiling in vacuum. **(b)** Specific absorption: copper wins. **(c)** The real trade-off: more torque means less thermal margin.

Copper wins on specific absorption by a factor 1.2 over graphene and 10^{10} over silica; graphene wins on limiting speed by a factor 26 over copper; silica wins on thermal margin and loses on everything else. There is no best material: there is the right material for the chosen channel.

For graphene the "absorption" entry needs care. A single sheet absorbs 2.3 % per pass ($\pi\alpha$), and a few-layer structure reaches an absorption comparable to copper; but in-plane thermal conductivity is 3000 W/m/K against 401 W/m/K, and the internal gradient, negligible for copper, must be computed here. V6 uses an effective density of 1300 kg/m^3 and an effective strength of 20 GPa: working values, not measurements, to be replaced with data from the individual sample.

9 Design levers

Table 9: Effect of each lever on the two figures of merit.

Lever	on Pe_r	on ω_{ss}	note
Power $\times 10$ ($1 \to 10\,\mu$W)	10 $\times$	10 $\times$	
Focusing NA 0.9 (w_0 500 $\to$ 361 nm)	1.92 $\times$	1.92 $\times$	
Linear polarization ($s = 0$)	0 $\times$	0 $\times$	
OAM $\ell = 5$ with $s = 1$ (mode overlap included)	1.05 $\times$	1.05 $\times$	
Compaction at equal mass ($R/10$)	1 $\times$	290 $\times$	torque unchanged; ω follows the drag ratio (10^2–10^3)
High vacuum ($p = 10^{-6}$ mbar)	1 $\times$	1.33×10^{10} $\times$	no true steady state: fracture becomes the limit
Material: graphene instead of copper	0.818 $\times$	0.818 $\times$	specific absorption changes; effect on $Pe_r = 0.82$

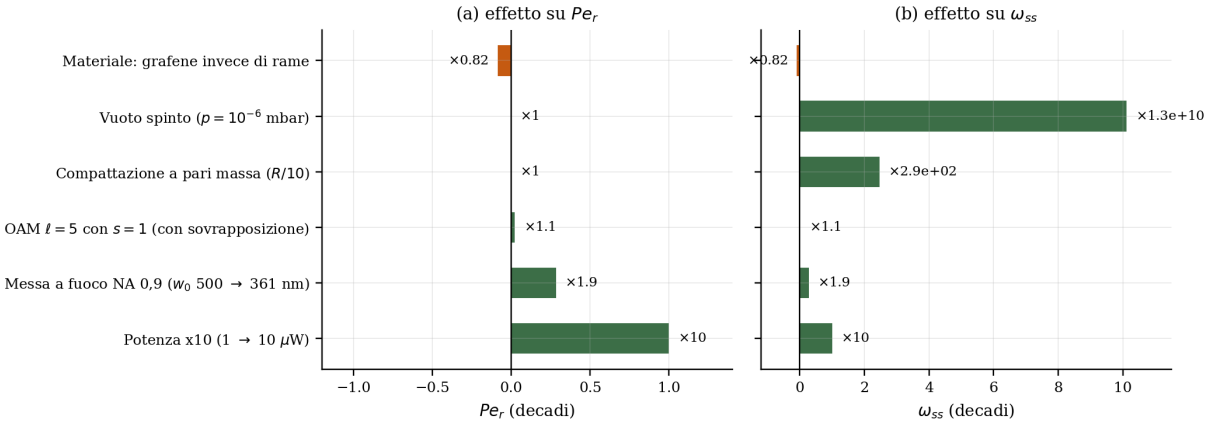


Figure 10: Effect of each lever on Pe_r (a) and ω_{ss} (b). The two columns do not coincide, and the difference is the message: compaction and vacuum change ω by ten decades and Pe_r by zero.

The most effective lever is power, and that is no surprise: $Pe_r \propto P$. The second is focusing, 1.92; the third is material, which however works against us (graphene absorbs 0.82 of copper at the same geometry). OAM, as §11 shows, is worth 1.05 and not 6: the table already includes the mode overlap, and this is the most important correction with respect to 6.0.

The two "free" levers do not exist. Polarization is necessary, not multiplicative: with $s = 0$ and $\ell = 0$ the torque is zero. Compaction is worth a factor 290 on ω_{ss} and zero on Pe_r: it improves the response without improving the drive, which is exactly what a sensor needs.

10 External check: silica at 1–6 GHz

This section does not claim a validation. It separates two statements that 6.0 kept together: order-of-magnitude compatibility on the *mechanical scale*, and validation of the *optical model*. The first is verified, the second is not, and no available datum supports it.

Table 10: Mechanical ceiling $f_{\mathrm{lim}} = (2\pi r)^{-1}\sqrt{\sigma_y/\rho}$ in GHz, for four radii and five values of the silica fracture strength.

σ_y (GPa)	$r = 25$ nm	$r = 50$ nm	$r = 75$ nm	$r = 100$ nm
0.5	3.03	1.52	1.01	0.759
1	4.29	2.15	1.43	1.07
3	7.43	3.72	2.48	1.86
5	9.6	4.8	3.2	2.4
8	12.1	6.07	4.05	3.03

Table 11: The experimental records and the effective strength they imply, $\sigma_y = \rho(2\pi r f)^2$ with $\rho = 2200\,\mathrm{kg/m^3}$.

f (GHz)	r (nm)	$v = 2\pi r f$ (m/s)	ρv^2 (GPa)
1	50	314	0.22
1	75	471	0.49
5	50	1571	5.43
6	50	1885	7.82

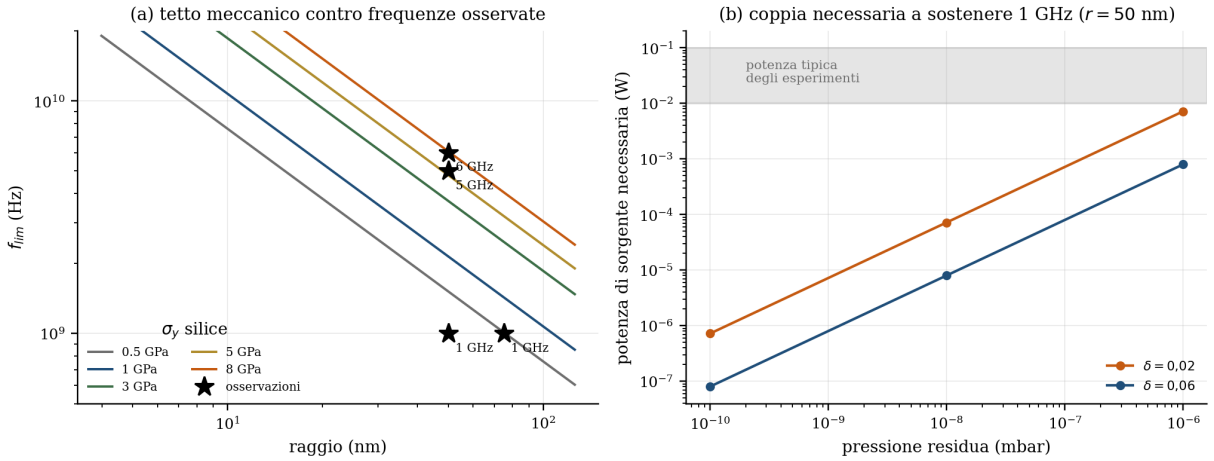


Figure 11: **(a)** Mechanical ceiling for five values of σ_y; the stars are the observed frequencies. Observations fall between 0.2 and 8 GPa, i.e. in the range of nanoscale silica. **(b)** Source power needed to sustain 1 GHz against molecular drag: far below the power used in the experiments, which serves the trap, not the torque.

(1) Mechanical compatibility. The reported (f, r) pairs imply σ_y between 0.22 GPa (1 GHz at 50 nm) and 7.8 GPa (6 GHz at 50 nm). The formula reproduces the observed frequencies for values of σ_y that are those expected for nanoscale silica, and this is a consistency check, not a prediction: σ_y was inferred from the observations, not measured independently. A single point with a chosen σ_y is not a validation; four points with σ_y free are a constraint.

(2) Torque is in surplus. At 1 GHz in vacuum at 1e-6 mbar the molecular drag on a 50 nm sphere is 1.0×10^{-27} N m; 0.8 mW of source power suffice with $\delta = 0.06$, and 8 µW at 1e-8 mbar, against the typical 10–100 mW. The difference is the optical trap. The model predicts surplus torque, which is consistent with the experiment, but does not prove it.

(3) What is missing for validation. One would need measured τ against P_{abs} at known geometry, and the dependence of ω_{ss} on β at variable medium. Neither is available for this device. The correct statement is therefore: the comparison offers a preliminary external check on

the mechanical scale, not an experimental validation of the Photonic Engine.

11 OAM as a sensitivity analysis

6.0 treated ℓ as a multiplier: $\tau = (\ell + s)P_{\mathrm{abs}}/\omega_{\mathrm{ott}}$, so $\ell = 5$ is worth six times. That is wrong at fixed *incident* power, because η_{int} does not stay constant.

For an $LG_{0\ell}$ beam of power P and waist w_0, the peak intensity, normalized to the Gaussian, is

$$g(\ell) = \frac{\ell^{\ell} e^{-\ell}}{\ell!}, \qquad g(1) = 0.368,\ g(5) = 0.176,\ g(10) = 0.125, \tag{6}$$

and on axis it vanishes for $\ell \geq 1$. A particle placed at the maximum intercepts σI_{peak}, so the net torque scales as $(\ell + s)\, g(\ell)$:

ℓ	0	1	5	10	20
$g(\ell)$	1	0.368	0.176	0.125	0.089
$\tau(\ell)/\tau(0)$	1	0.74	1.05	1.38	1.87

The gain is of order $\sqrt{\ell}$, not ℓ, and for $\ell = 1$ it is even a loss. There is, however, a practical advantage that V6 records and does not quantify: a beam with $\ell \neq 0$ carries angular momentum in an annular region, which allows the trap light to pass through the centre and the torque light on the ring. That is a design argument, not a physical one, and it must be verified by measuring $\eta_{\mathrm{int}}(\ell)$.

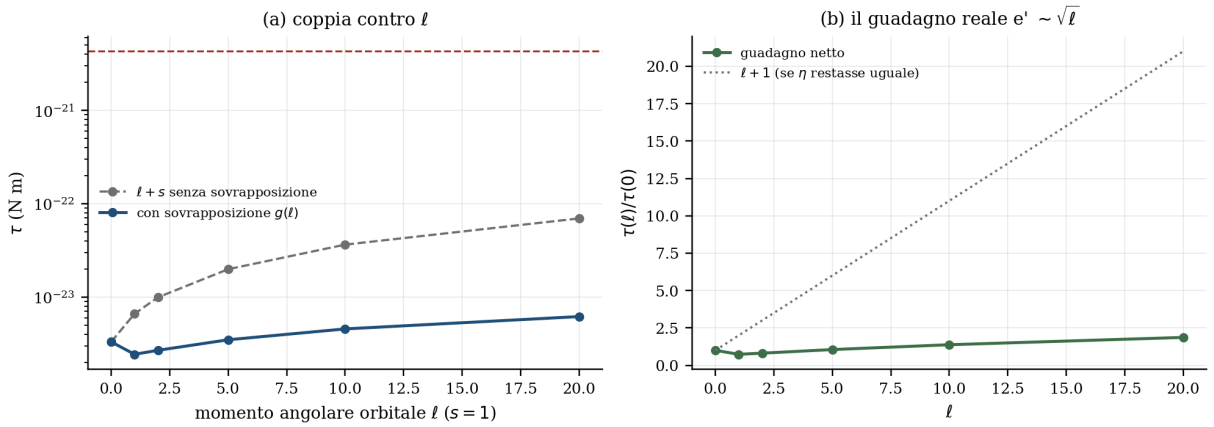


Figure 12: **(a)** Torque against ℓ with and without the mode overlap. **(b)** Net gain: the grey curve is what one would obtain if η stayed constant, the green one is what is obtained.

12 Limits, falsifiability, protocol

12.1 What V6 has not done

1. **Full-wave calculation of the whole frame.** The superposition is incoherent and declared; a DDA/FDTD run can change σ_{pr} by a factor 4. It is the first task, and it conditions the entire torque section.
2. **Per-shape optics.** Disc, ring and dumbbell have a band, not a value. Without DDA their Pe_r stays conditional.

3. **Thermo-mechanical coupling.** Temperature is computed in steady state on a rigid body; differential expansion between core and frame and the resulting loss of planarity are not treated.
4. **Counter-spin dynamics.** The 45/376 ratio is treated as a kinematic constraint; the dynamics of that constraint are not computed.
5. **Real fluid.** Newtonian water at 20 °C; a biological medium is viscoelastic.

12.2 How to refute V6

1. **Torque against absorbed power.** Equation (2) predicts $\tau/P_{\mathrm{abs}} = 1/\omega_{\mathrm{ott}} = 2.82\times10^{-16}$ s. If a measurement on a 40 nm copper particle exceeds it, the model is incomplete and the scattering channel transfers angular momentum coherently.
2. **Dependence on β.** Identity (3) predicts that Pe_r does not change when going from water to water-glycol while ω_{ss} scales as $1/\beta$. If Pe_r changes, the drag model is wrong.
3. **Thermal threshold in vacuum.** V6 predicts melting at 0.13 –0.66 µW. If melting occurs at much higher power, the radiative balance is incomplete (conduction through the levitation medium) or the area is much larger than the geometric one.

12.3 Protocol for the full-wave audit

- Geometry: exact V5 frame, wires 6.4 / 15 / 9 nm, core 40 nm.
- Material: copper with Drude $\varepsilon(a)$, plus a Cu_2O shell of 0, 1, 2 nm.
- Discretisation: step $\leq \lambda/20$ in the metal (≤ 13 nm), with two-step convergence check.
- Incidence: four orientations, circular polarization.
- Output: σ_{ext}, σ_{abs}, g, σ_{pr}, and the comparison with the incoherent sum used here.

A The 720° cycle and the 45/376 ratio

This material should be kept, not promoted. Three statements, in decreasing order of strength.

First, solid. The V5 scene completes exactly 720° on the Spin and 86.1702° on the Counter-spin in 240 frames, with absolute keyframes and no reduction modulo 360. It is a verifiable fact in the `.blend` file and in the code.

Second, conventional. The double cover of $SO(3)$ is a property of $SU(2)$: $U(\theta) = \exp(-i\theta\sigma_z/2)$ changes sign at 2π and returns to $+1$ at 4π. A rotor is a macroscopic classical body and its geometric periodicity is 2π; the 4π periodicity is a construction choice of the animation, not a measurable signature.

Third, not to be used. The ratio $I_2/I_1 = 45/376$ closes the kinematics, not the dynamics. V5 achieved closure by adding 3.06×10^{-20} kg at 42 nm from the centre (2.9×10^5 copper atoms, 1.1 % of the frame mass). V6 does not use that mass: the point of application falls inside the core and corresponds to no element of the scene. Above all, nothing guarantees that the optical and dissipative torques spontaneously produce that ratio. A rational ratio between two angular velocities guarantees geometric repetition *if* the ratio is maintained; maintenance requires a mechanical constraint, and a mechanical constraint between two rotors makes them a single rigid body with a single moment of inertia. The Counter-spin cannot be at the same time independent (to have a ratio of its own) and constrained (to keep it).

The ratio 45/376 remains a property of the shape, invariant over nine orders of magnitude in scale and across the three materials. As such it should be cited, without the balancing mass and without the quantum aura that V4 attributed to it.

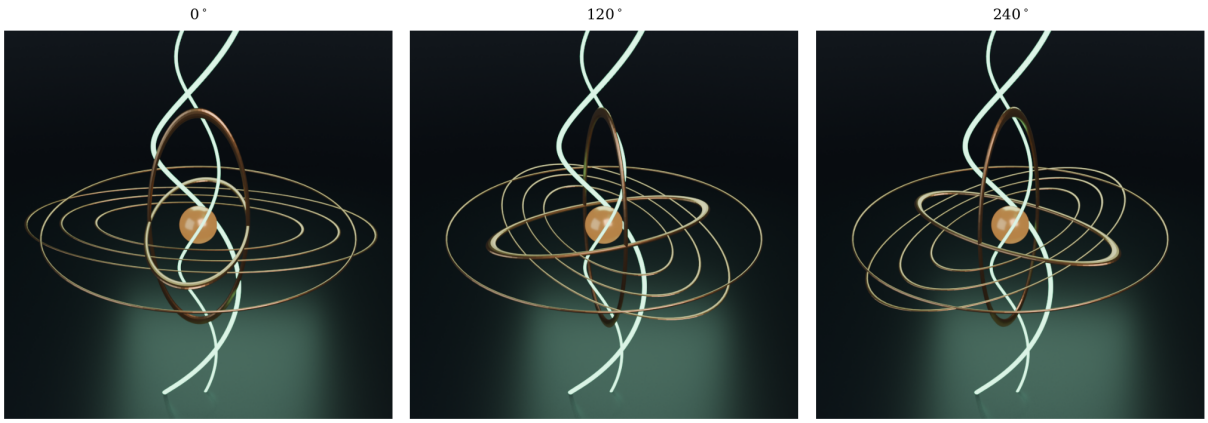


Figure 13: Three orientations of the scene at 0, 120 and 240°. The animation is a check of geometric consistency of the cycle, not a physical simulation.

B Excluded material and its destination

Two themes of V5 do not enter V6, and this is not an oversight. The *ark* and the *blood* — cosmological metaphors and the reference to a biological medium as a justification for the design — belong to a different discourse, and mixing them with a dimensional analysis weakens both: they belong in a separate text. *Pulsed operation* remains, but demoted to a thermal lever (§5): the mechanical time constant of 1.1 ns makes it impossible for pulsing to change the dynamics.

C Code, data and reproducibility

Every number in the text comes from one of four scripts, none of which contains hard-coded data.

file	function
`v6_nucleo.py`	optics (Mie, cylinder, core-shell), drag, thermal, dynamics; self-test
`v6_studi.py`	the nine studies, JSON in `dati/`, tables in `tabelle/`
`v6_figure.py`	the twelve figures from the computed records
`v6_blender.py`	the scene and the renders in `blender/`

The self-test checks convergence of the cylindrical series against Rayleigh, the conservation $Q_{\mathrm{ext}} \geq Q_{\mathrm{sca}}$, identity (3) and the limits of the coated sphere. Anyone wishing to contest a number must contest one of these checks: that is the point of a revision.

Acknowledgements

This revision stems from an external critique that indicated the residuals, the five-phase structure, the four-question criterion and, in 6.1, the five material errors corrected here: wire radii, the number 0.119667646, the radiating area, the circularity of the baseline and the single point in the GHz comparison. V6 is an execution of that programme.

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